

Data Centre Management (DCM) System User Guide

System Access & Security

Purpose: Outlines the security protocols governing user access and session management.

Session Rules:

- Single Session: Users can only be logged into one session at a time. If you log in from a new location, your previous session will be automatically terminated.
- Token Expiry: Authentication tokens are valid for 8 hours. After this period, the session expires, and you must log in again.

Late Access Rule:

- If you are late and possess the privilege to update the checklist, you can access the checklist page without logging in, as these tasks must be completed before 12:00 AM.
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Dashboard

Purpose: The Dashboard serves as your central command center, providing a real-time summary of critical operations and items requiring immediate attention.

How to use the Dashboard:

1. Review the dashboard cards to see summaries of pending work, open observations, and today's roster.
2. Click directly on any dashboard card to navigate to that specific module for detailed information.
3. Check the "Announcements" section for important organizational updates.

System Rules: Access to the Dashboard is restricted to users with the "View Dashboard" privilege.

User Management

Users

Purpose: This module is used by authorized administrators to manage the lifecycle of user accounts, from creation to deactivation.

How to manage users:

1. To view users, navigate to the Users module; your view is filtered based on whether you have "View All" or "View Department" privileges.
2. To create a user, click "Create User," enter the required profile details, and assign appropriate roles.

- 3. To update an account, select a user from the list, click "Update," and save your changes.
- 4. To remove access, use the "Inactive" or "Delete" function for users who have left the organization.

Important Notes:

- Replacement Rule: When adding a user who is replacing a former employee, link the new profile to the inactive record of the predecessor if prompted.
 - Access Rule: Inactive users are completely barred from system operations and application login.
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Roles

Purpose: Roles allow you to group specific privileges together, ensuring users have the exact level of access needed for their job functions.

How to manage roles:

- 1. Navigate to the Roles module and click "Create Role" to define a new permission set.
- 2. Select and assign multiple privileges to the role from the available list of module permissions.
- 3. Use "Update Role" to modify the privilege list for an existing role at any time.

System Rule: Every feature in the DCM system is gated by privileges; users cannot see or use tools without an assigned role containing those specific permissions.

Work Management

Works

Purpose: The Works module is designed to track, assign, and manage the progress of various operational tasks within the data center.

How to manage work tasks:

- 1. To start a task, click "Create Work," enter the task details, and assign it to a specific user.
- 2. As an assigned user, open your task and use the comments section to provide status updates and remarks.
- 3. To hand over a task, use the "Transfer" feature to assign the item to a user in the next shift.
- 4. Once work is finished, update the status to reflect the task is complete.

System Rules:

- Priority Ranking: Tasks must be categorized as Low, Medium, or High priority.
 - Workflow Rule: High-priority items must always be addressed before lower-priority tasks.
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Department Management

Departments

Privileges:

- View Department
- Create Department
- Update Department
- Delete Department

This module is used to create and manage departments within the organization.

Roster Management

Roster

Purpose: The Roster module coordinates staff availability and shift schedules, ensuring adequate coverage for data center operations.

How to manage the roster:

- 1. Create a Roster: Depending on your assigned privileges, click "Create Roster" to begin building the duty schedule for a specific period.
- 2. Allocate Shifts: Assign staff members to specific shifts and utilize the "Roster Split-up" tab to plan weekly duty allocations against monthly targets.
- 3. Visibility: Once created, the roster is immediately visible to all users within the department, regardless of its approval status.
- 4. Finalize & Approve: Once the schedule is finalized, the Department Head should click "Approve Roster" to provide official confirmation. Users can view their shifts in the "Duty Roster" tab.
- 5. Update & Modify: If changes are needed, authorized users can use "Update Roster." Please note that any modification will reset the approval status, requiring the Department Head to re-approve the schedule.

Important Notes:

- Roster Reset: Any edit to an approved roster automatically forces it back to "Pending Approval" status.
- Automatic Leave: Users who are scheduled but do not log in are automatically flagged as being on leave by the system.
- Conflict Rule: The system prevents assigning a user to more than one shift per day or assigning an early shift (Shift 1/2) immediately following a night shift (Shift 3).

Printing

The roster can be printed directly from this module.

Observation Management

Observations

Privileges:

- View Observations
- Create Observations
- Update Observations
- Delete Observations
- Create Observation Categories
- Update Observation Categories
- Delete Observation Categories

The Observations module is used to record and track operational observations.

Features

- Create observation records.
- Create and manage observation categories.
- Filter observations using categories.
- Mark observations as resolved.

Users can mark observations as resolved by selecting the corresponding observation and updating its status.

Visibility

Observations are visible to all users within the same department.

Observation Checklist

When creating an observation, users must complete a checklist to ensure the issue has been reported to the responsible personnel.

Inventory Management

Inventory

Privileges:

- View All Inventory
- View Department Inventory
- Create Inventory
- Update Inventory
- Delete Inventory

The Inventory module helps track and manage equipment and consumables.

Features

- Search inventory items.
- Track quantities.
- Maintain rack and drawer locations.
- Update stock movements.

Inventory must be updated whenever items are issued or returned.

Items issued through Hardware Issuance Requests are automatically deducted from inventory.

Inventory Types

General Inventory

Tracks standard inventory items.

Returnable Inventory

Tracks items that must be returned after use.

For returnable inventory:

- Record the recipient's name.
- Record issue date and time.
- Record return date and time.
- Maintain complete issue history.

Issued items are automatically marked as unavailable until returned.

Configuration Management

Configurations

Privileges:

- View Configurations
- Update Configurations
- Delete Configurations
- View Server Details
- Create Server Details
- Update Server Details
- Delete Server Details
- View Cluster
- Create Cluster
- Update Cluster
- Delete Cluster

The Configuration module is primarily intended for administrators.

Available Sections

- Server Masters
 - Request Configurations
 - Attendance
 - BMS Checklist
 - Cluster Checklist
 - Morning Checklist
 - Notifications
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Server Masters

Maintains reusable master data such as:

- Cluster Types
- Hypervisors
- Server Models

These masters ensure consistency when creating server records.

Request Configuration

Request workflows are configured here.

The system automatically assigns requests and updates statuses according to configured workflows.

How to manage request stages:

- In the Request Configurations section, you can define the workflow order.
- Click the "Up" or "Down" arrow buttons next to each stage to rearrange the sequence as required by your business process.

Status Types

Functional Statuses

Functional statuses contain specialized fields and business logic.

Non-Functional Statuses

Non-functional statuses are standard workflow statuses used for routing and assignment.

Both status types influence request flow.

Supported Functional Statuses

VM Creation Requests

Status	Purpose
IP Issuance	Assign IP address
Cluster Deciding	Select cluster and node
VM Creation	Confirm VM creation and monitoring registration
VM Backup	Specify backup location

Data Centre Entry Requests

Status	Purpose
Mark Entry Time	Record visitor entry time
Mark Exit Time	Record visitor exit time

Status matching is case-insensitive and supports partial keyword matching.

Attendance Configuration

Administrators can configure:

- Cycle Start Day
- Cycle End Day
- Standard Shift Start Time
- Late Grace Period
- Maximum Allowed Days
- Attendance Tracking Roles
- Shift Timings
- Roster Row Assignments

Late Login Restriction

Users logging in after:

Shift Start Time + Grace Period

are considered late.

Such users cannot access the application until approval is granted by their manager.

Checklist Configuration

Administrators can create, update, and delete fields for:

- BMS Checklist
 - Cluster Checklist
 - Morning Checklist
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Notification Configuration

Used to configure notification types and define which roles receive specific notifications.

Server Management

Server Details

Privileges:

- View Server Details
- Create Server Details
- Update Server Details
- Delete Server Details
- View Cluster
- Create Cluster
- Update Cluster
- Delete Cluster

This module is used to maintain physical and virtual server information.

Request Management

Requests

Request privileges are controlled through configured workflows and assigned roles.

Features

- Create requests.
- Route requests automatically.
- Return requests for rework.
- Track approval stages.

Request Types

New request types can be created through the Request Types section.

Automatic Actions

VM Creation Request

Upon completion:

- VM is automatically added to the server inventory.

Data Centre Entry Request

Upon completion:

- Entry is added to Visitor Logs.

Hardware Issue Request

Upon completion:

- Inventory quantity is automatically updated.

Global Search

The Search module provides a centralized search interface.

Users can search for:

- Virtual Machines
 - Clusters
 - Nodes
 - Racks
 - Physical Servers
 - Inventory Items
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Monitoring

Server Monitoring via vCenter

Status: In Progress

Will provide direct integration with VMware vCenter for infrastructure monitoring.

Server Monitoring via Ping

Allows monitoring of:

- Servers
 - Applications
 - Network availability
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Attendance Management

How to validate attendance:

- The system automatically validates attendance by comparing application login records against official attendance records captured at the facility gate.
- If discrepancies occur, use the "Regularization" feature to request corrections, subject to manager approval.

Attendance Calculations

The system calculates:

- Worked hours
- Login status
- Attendance compliance

If a user does not log out properly, worked hours may not be captured accurately.

Regularization

Regularization allows users to request corrections for attendance records when exceptional circumstances occur.

Examples:

- Late arrival
- Early departure
- Emergency situations

Manager approval is required before regularization is applied.

Documentation

The Documentation module stores departmental documents.

Uploaded documents are visible to all users within the department.

Audit Logs

Administrators can view system activity logs and API usage records.

GET requests are excluded from audit logging.

Daily Activities

Daily Activities are used to manage operational checklists.

Supported Checklists

- BMS Checklist
- Cluster Checklist
- Morning Checklist

Rules

Checklists can only be created on the current date.

Users cannot create:

- Yesterday's checklist
- Future checklists

Checklist entries must be completed before midnight.

Once marked as completed, checklist entries cannot be edited.

Visitor Logs

The Visitor Logs module records all visitor entry and exit activities.

Periodic Activities

Periodic Activities help track:

- AMC schedules
- Recurring maintenance
- Scheduled operational activities

Notifications are generated for upcoming activities.

Announcements

The Announcements module is used to publish organizational announcements.

Announcements are displayed on the dashboard.

Telephone Directory

Purpose: Central repository for storing and retrieving contact numbers.

How to use:

- Navigate to the Telephone Directory.
 - Search for contacts by name or department to quickly retrieve necessary telephone numbers.
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IP Management

Purpose: Monitor and manage the distribution of IP addresses.

How to use:

- Navigate to the IP List module to view a real-time status of all IP addresses.
 - The interface distinguishes between "Used" and "Free" IPs, allowing for efficient allocation and network planning.
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Notifications

The Notifications module is responsible for generating and delivering system notifications based on configured events and workflows.

Operational Logs

Operational Logs are used by staff members to record operational updates and shift handover information, ensuring continuity of work across shifts.